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High Performance Processors and Systems

April 30, 2013

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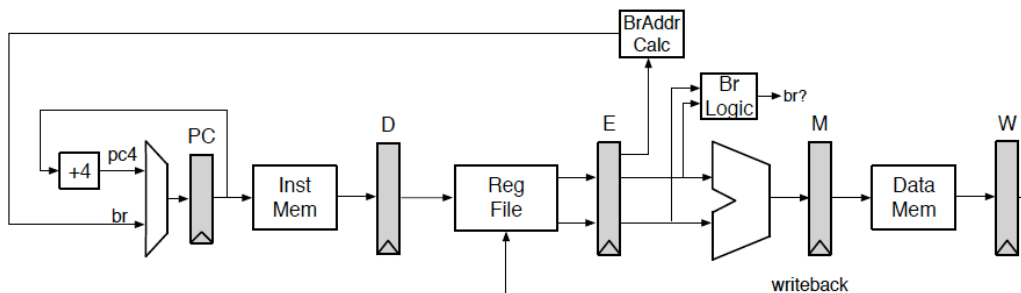
Name
Last Name

Problem 1 (6 points)	
Problem 2 (6 points)	
Problem 3 (6 points)	
Problem 4 (6 points)	
Problem 5 (6 points)	
Total (30 points)	

To pass the exam (the following conditions are in AND):

- To get access to the 2nd midterm the overall score has to be equal or higher than 14
- At least 1 problem has to be completely correct

Problem 1



Consider the above 5 stages pipelined architecture.
Indicate in the following table what kind of hazards can occur in this pipeline, motivating your answer

Structural hazards	
Data hazards	
Control hazards	

Consider the following executed code.

0x1000: ADDI R4, R0, R1
 0x1004: ADDI R5, R0, R2
 0x1008: BEQ R4, R5, 0x1000
 0x100C: LW R7, 4(R6)
 0x1010: OR R5, R7, R5
 0x1014: SW R5, 8(R6)

Complete the table and define how many clock cycles are needed.

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18
ADDI R4, R0, R1	F	D	E	M	W													
ADDI R5, R0, R2		F	D	E	M	W												
BEQ R4, R5, 0x1000			F	D	stall	stall	D	E	M	W								
LW R7, 4(R6)				F				D	E	M	W							
OR R5, R7, R5								F	D	stall	stall	D	E	M	W			
SW R5, 8(R6)									F	stall	stall	stall	D	stall	stall	E	M	W

Provide the architectural solutions to improve the performance for this code for the data hazards, motivating your answers and compute the performance improvement you can obtain.

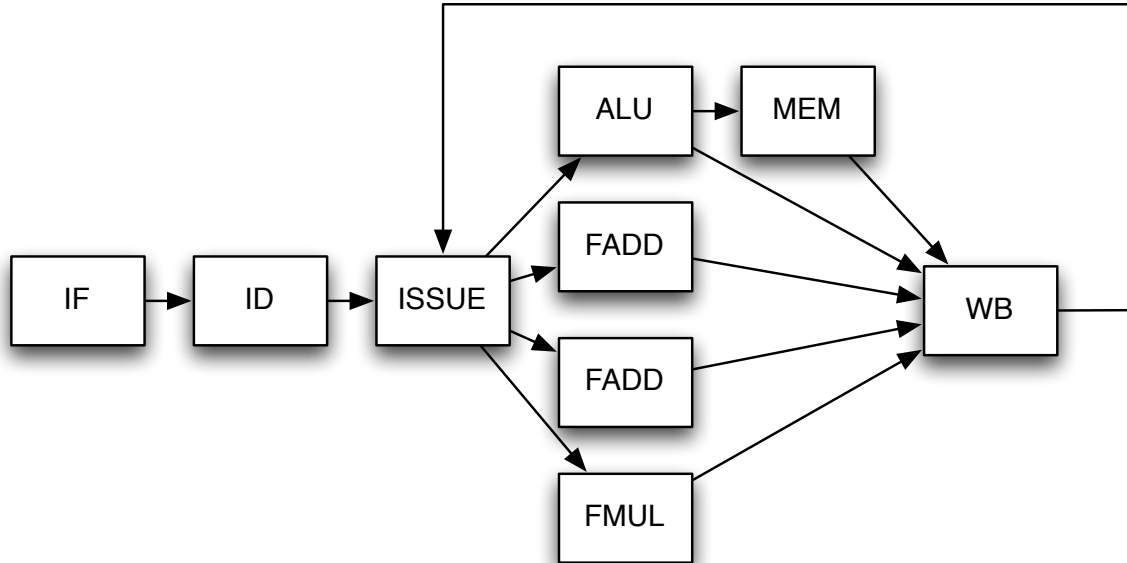
Considering the control hazard, what improvement can you provide with static branch prediction?

How can you provide precise exceptions in this pipeline?

Answer

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

All functional units are pipelined

ALU operations take 1 cycle

Memory operations take 2 cycles (includes time in ALU)

Floating-point add instructions take 3 cycles

Floating-point multiply instructions take 5 cycles

There is no register renaming

Instructions are fetched, decoded and issued in order

The issue stage is a buffer of unlimited length that holds instructions waiting to start execution

An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard

Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

I - When an instruction enters the issue stage

E - When an instruction starts execution (enters FU)

C - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

I ₁	L.D	F0, 0(R0)
I ₂	MUL.D	F0, F1, F0
I ₃	ADD.D	F2, F1, F1
I ₄	ADD.D	F0, F3, F3
I ₅	ADDI	R1, R3, 8
I ₆	ADD.D	F3, F0, F0

Problem 1.A

Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

[illegible]

Problem 3

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 100, R1 is set to 0)

```
LOOP:    ADDI    R1    R1    300
          LD      F1    0     R0
          ADDD    F2    F1    F1
LOOP2:    MULTD   F2    F3    F1
          ADD     F4    F1    F1
          SUBI    R1    R1    1
          BNEZ    R1    LOOP2
          SUBI    R0    R0    1
          BNEZ    R0    LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 4

Consider a set associative cache. Complete the following table marking whether the following modifications to the cache will impact τ by increasing or decreasing each of the three terms composing the memory performance equation.

Memory access time =

Double the associativity (capacity and line size constant)			
Halving the line size (associativity constant)			
Doubling the number of sets (capacity and line size constant)			

Problem 5

Assume that the following code is executed on a CPU with SCOREBOARD and with the following units:

- 1 LOAD/STORE unit with Latency= 1
- 2 MULT units with latency = 10
- 1 DIVIDE unit with latency = 30
- 1 ADD/SUBD unit with latency 2

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	2	3	4	5
MULTD F0 F4 F2	3	6	16	17
ADD F8 F2 F6	4	6	8	9
DIVD F12 F0 F6	5	18	46	47
SUBD F6 F8 F2	6	7	9	10

Is the Pipeline schema correct?
If no, explain in details all the problems.

Answer

